

Dietary sources of inorganic microparticles and their intake in healthy subjects and patients with Crohn's disease.



Dietary microparticles are non-biological, bacterial-sized particles. Endogenous sources are derived from intestinal Ca and phosphate secretion. Exogenous sources are mainly titanium dioxide (TiO₂) and mixed silicates (Psil); they are resistant to degradation and accumulate in human Peyer's patch macrophages and there is some evidence that they exacerbate inflammation in Crohn's disease (CD). However, whether their intake differs between those with and without CD has not been studied. We aimed to identify dietary microparticle sources and intakes in subjects with and without CD. Patients with inactive CD and matched general practice-based controls (ninety-one per group) completed 7 d food diaries. Intake data for dietary fibre and sucrose were compared as positive controls. All foods, pharmaceuticals and toothpastes were examined for microparticle content, and intakes of Ca and exogenous microparticles were compared between the two groups. Dietary intakes were significantly different between cases and controls for dietary fibre (12 (SD 5) v. 14 (SD 5) g/d; P=0.001) and sucrose (52 (SD 27) v. 45 (SD 18) g/d; P=0.04) but not for Ca. Estimated median TiO₂ and Psil intakes (2.5 and 35 mg/individual per d respectively, totalling 10(12)-10(13) microparticles/individual per d) were broadly similar to per capita estimates and while there was wide variation in intakes between individuals there was no significant difference between subjects with CD and controls. Hence, if exposure to microparticles is associated with the inflammation of CD, then the present study rules out excess intake as the problem. Nonetheless, microparticle-containing foods have now been identified which allows a low-microparticle diet to be further assessed in CD.